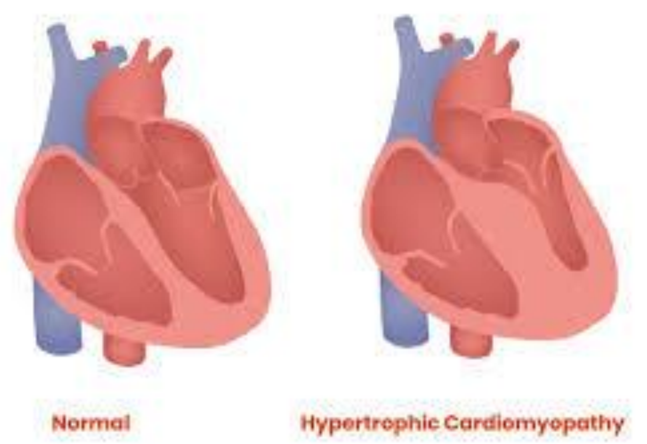
Reducing the Severity of Cardiac Hypertrophy in *Drosophila melanogaster* with Cholecalciferol

TMED

Background

Cardiac Hypertrophy

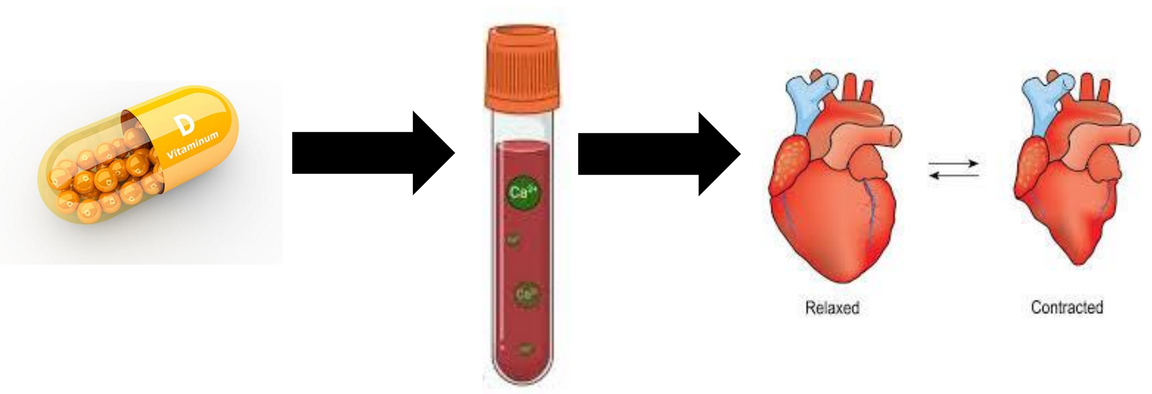


https://pennstatehealthnews.org/2023/08/the-medical-minute-understanding-hypertrophic-cardiomyopathy/

Notable Changes once a Heart become Hypertrophic:

- Thickening of the Heart Muscle
- Increased Actin and Myosin Production
- Impaired Relaxation
- Increased risk of Heart Failure and Blood Clots

Vitamin D Supplementation



https://www.ezmedlearning.com/blog/cardiac-cycle

Vitamin D supplementation increases calcium absorption in the small intestine leading to higher calcium levels in the blood stream.

Calcium increases the frequency and force of heart contractions by causing myosin to bind with actin filaments in the heart muscle.

Problem

Cardiac Hypertrophy is a heart disease that affects 1.7 million Americans. Currently there is no effective treatment for Cardiac Hypertrophy, only medications to suppress its symptoms. The issue with these medications is that they can not cover all symptoms, and are expensive, and some are dangerous.

1 in 4 people diagnosed with cardiac hypertrophy dies within 5 years of the diagnosis.

Variables/Hypothesis

If Cholecalciferol is incorporated into the diets of *Drosophila melanogaster* induced with Cardiac Hypertrophy, then the severity of Cardiac Hypertrophy in the flies will decrease proportionally to the concentrations of their diets.

IV: Concentration of Cholecalciferol (Vitamin D3) added into the diets of fruit flies.

DV: The Band Intensity of Actin

Controls: Wild type *Drosophila melanogaster,* UAS-RAS85DV12 x Tin C – Gal 4 *Drosophila melanogaster* not treated with cholecalciferol.

Constants: Temperature, humidity, age of flies, strain of flies used, food ingredients besides cholecalciferol.

Procedure

Treatment Method

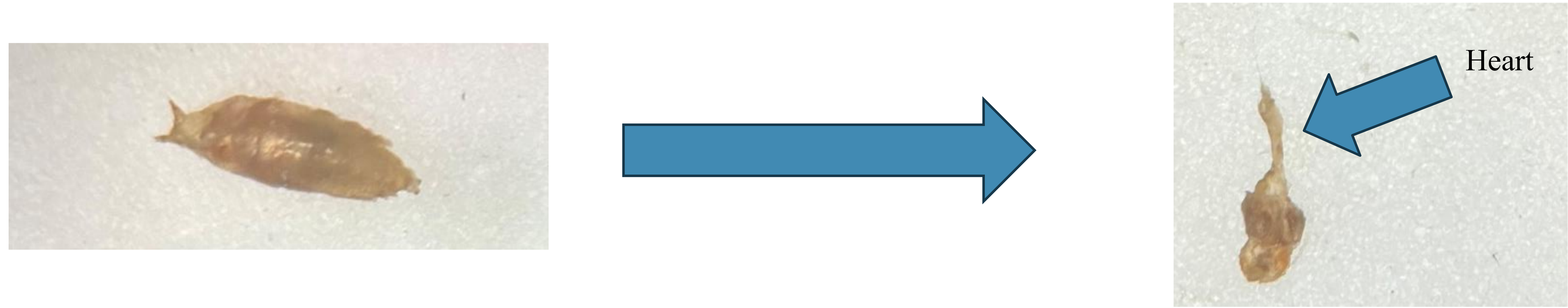
Method for Creating and Treating Flies with Cardiac Hypertrophy

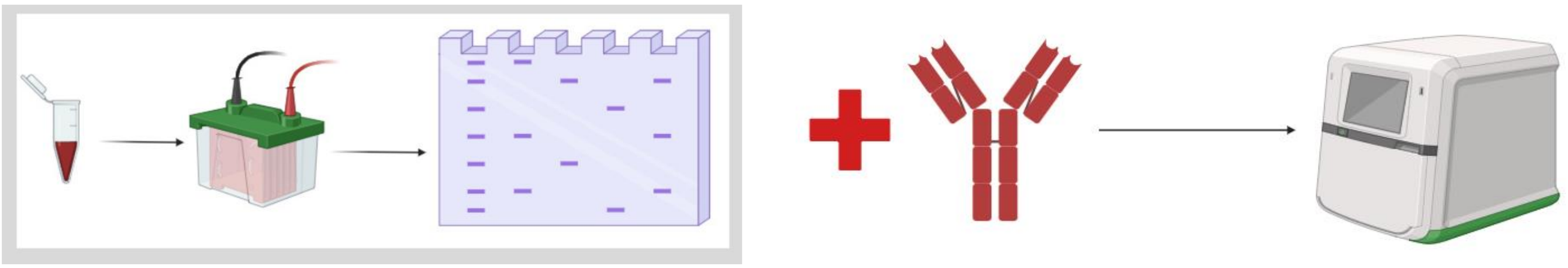
1. Incorporate cholecalciferol into fly food so that there is vials with concentrations of 12.5, 25, and 50 micromolar.
2. Cross a UAS-Ras85DV12 stock with a TinC-Gal4 stock to form a cross that has Cardiac Hypertrophy.
3. Insert 10 or more flies with Cardiac Hypertrophy into a vial with each concentration.
4. Wait for the flies to reproduce, as their offspring will show the effect of treatment then western blot.

	Normal Diet	Cholecalciferol Diet
No Cardiac Hypertrophy	Negative Control (*Drosophila melanogaster*)	Condition Control
Cardiac Hypertrophy	Positive Control (*UAS-Ras85D^{V12} x Tin C-Gal4*)	Experimental Group

Figure 1: Table depicting the different control groups and experimental group

Western Blot Analysis





Graphic made in Bio render

1. Dissect the Hearts of *Drosophila* Larva as their hearts will show the effects of the diets
2. Breakdown the Heart tissue with Rippa Buffer and extract the proteins
3. Separate by Protein size using Gel Electrophoresis.
4. Stain with anti-actin or anti-myosin antibody.
5. Take pictures with the ChemiDoc imaging system

Results

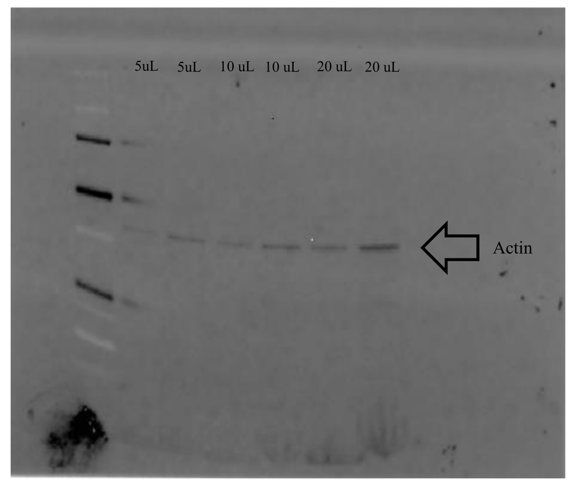
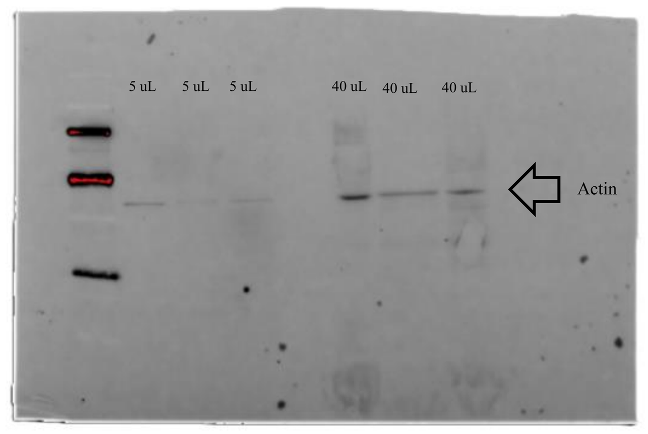


Figure 2, Image taken by Chemidoc of a Western Blot of Actin in Drosophila Hearts, Trial 1



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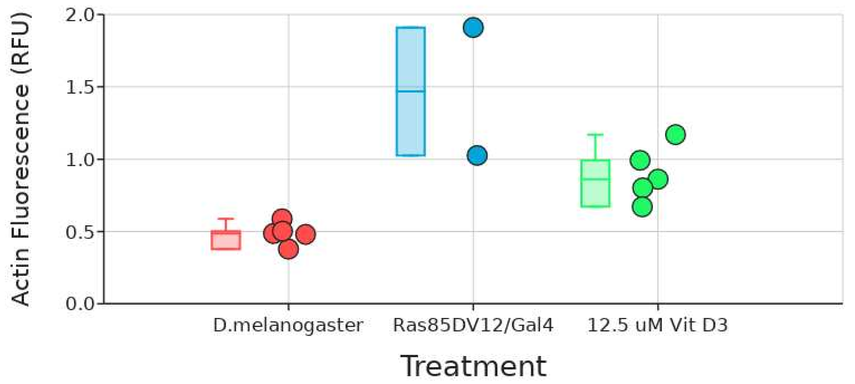

Note: Box and whiskers plot shows the median value (line), interquartile range (box), and the extent of the data (whiskers above and below at min / max data points).

Figure 4: Graph depicting the Actin Fluorescence of each treatment group.

Effect	Degrees of Freedom (df)	F-statistic (MS / MS residual)	P-value	Interpretation of P Explain...
Group	2	11	<0.01	A P-value of <0.01 means **extremely strong confidence** that the groups are different.
Error or Residual	9			

Results of post-hoc Tukey's Range Test:

Pair	Mean diff	P-value	Interpretation of P
12.5 um - Negative Control	0.412	0.07	Means: **some confidence** that the groups are different
Positive Control - Negative Control	0.981	<0.01	Means: **extremely strong confidence** that the groups are different
Positive Control - 12.5 um	0.569	0.06	Means: **some confidence** that the groups are different

Figure 5: Results of a T-Test and Post-Hoc test based on the groups in Figure 4.

Discussion

As only two trials has been completed, and one trial with data from all three groups this study's conclusions are limited. However, it is worth noting that there is no statistical difference between flies without Cardiac Hypertrophy and those with Cardiac Hypertrophy but treated with 12.5 micromolar of Cholecalciferol. The group with Cardiac Hypertrophy still did have more actin than the negative control. There was also no statistical difference between flies with and without treatment. However, the flies with HCM but without treatment had higher actin levels than those with HCM but treated with 12.5 uM dosage of cholecalciferol. Also, there was a statistical difference between the positive control flies and the negative control flies.

Since there was no statistical difference between the experimental group and the positive control, this indicates that cholecalciferol treatment seemingly did not reduce the severity of Cardiac Hypertrophy. The hearts did not statistically change; however, the amount of actin fluorescence was lower in the experimental group then the positive control. Also, the post-hoc test between the experimental group and the positive did return a p-value of 0.06 which is just barley greater the significance level of 0.05. However, based on the limited collected, there is no statistical proof that cholecalciferol decreased the severity of HCM. The p–value of less than 0.01 in the post hoc test comparing the negative control and positive control signifies that the cross did work and the experimental flies did develop cardiac hypertrophy.

Future Work

In the future more trials will have to be completed to gain a more complete understanding of what the Calciferol treatment does. This includes more trails of what has already been done, as well as the inclusion of more groups in data collection. Data should be collected on the effects of varying concentrations of cholecalciferol treatment to see if a different dosage yields more significant results than a concentration of 12.5 uM of cholecalciferol.

References

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